

PORTALEGRE, Portugal

WMO: 085710

Lat: 39.2942N

Lon: 7.4214W

Elev: 1961

StdP: 13.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.7	36.9	13.7	11.9	46.1	18.7	15.2	47.9	23.4	44.6	21.2	45.1	10.4	320	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.4	96.3	65.3	93.2	64.5	90.1	63.6	67.9	88.8	66.6	86.3	65.5	84.3	5.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.3	90.3	68.9	60.9	86.0	68.1	59.7	82.2	67.7	33.2	89.0	32.2	86.5	31.3	84.5	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	17.8	15.8	DB	31.0	101.7	2.8	2.4	29.0	103.4	27.4	104.8	25.8	106.2	23.8	107.9	(4)
			WB	26.9	69.5	3.4	1.7	24.4	70.7	22.4	71.7	20.5	72.6	18.0	73.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	47.5	49.1	53.2	55.8	62.5	70.8	75.1	75.9	70.7	62.7	52.8	49.1
	DBStd	12.31	4.99	5.62	6.79	7.74	8.49	8.65	7.61	7.32	7.45	7.20	5.84	4.71
	HDD50	360	106	77	42	23	2	0	0	0	0	2	35	73
	HDD65	2865	542	444	370	295	155	36	4	2	26	131	368	492
	CDD50	4193	29	53	140	197	389	623	778	804	621	396	118	45
	CDD65	1227	0	0	4	19	77	209	318	341	198	60	1	0
	CDH74	12223	0	0	5	73	592	2185	3660	3772	1651	285	0	0
	CDH80	5248	0	0	0	3	162	913	1734	1783	592	61	0	0
Wind	WSAvg	7.4	7.3	7.8	8.2	7.9	7.4	7.2	7.3	7.1	6.5	6.9	7.7	7.6
Precipitation	PrecAvg	34.7	4.8	4.2	3.0	3.1	2.5	1.4	0.3	0.3	1.7	3.9	4.7	4.8
	PrecMax	57.0	17.2	10.5	11.0	11.5	8.3	7.2	2.3	1.9	4.6	11.7	15.2	15.4
	PrecMin	23.7	0.0	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.5
	PrecStd	8.2	3.9	3.1	2.6	2.2	1.9	1.5	0.4	0.4	1.3	2.9	3.7	3.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	66.1	73.9	80.2	89.5	97.9	99.7	100.2	94.3	86.3	70.1	63.9
		MCWB	50.3	52.6	54.9	59.7	62.4	66.3	66.3	66.2	64.2	59.6	57.0	52.6
	2%	DB	58.7	62.6	70.0	75.9	84.4	92.8	96.2	96.3	90.2	80.8	66.2	60.2
		MCWB	48.2	51.6	53.6	58.2	62.3	64.5	65.3	65.3	62.7	58.7	55.2	51.0
	5%	DB	56.1	59.8	67.0	72.0	80.4	89.2	92.9	92.9	86.6	76.8	63.3	57.5
		MCWB	48.1	49.7	52.5	56.3	60.4	63.9	64.5	64.5	61.8	57.6	53.9	50.4
	10%	DB	54.0	57.1	63.7	68.6	76.7	85.3	89.4	89.3	82.9	73.0	60.6	55.4
		MCWB	47.8	48.4	51.0	54.6	59.0	62.6	63.5	63.4	60.8	56.9	53.0	50.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	56.1	58.2	62.1	66.5	68.9	70.1	69.3	66.7	64.8	61.2	56.5
		MCDB	56.6	61.7	69.0	75.6	82.8	90.2	91.9	92.7	85.5	73.0	64.1	57.5
	2%	WB	53.0	53.7	55.8	59.5	64.2	66.9	67.8	67.6	65.2	62.6	58.9	54.8
		MCDB	54.7	57.7	64.7	72.7	79.8	86.9	89.0	88.8	81.9	71.0	61.9	56.6
	5%	WB	51.5	52.1	54.2	57.6	62.3	65.4	66.3	66.3	64.0	61.2	57.1	53.2
		MCDB	53.2	56.5	62.1	69.7	77.1	84.4	86.3	86.5	79.5	69.2	60.8	55.3
	10%	WB	50.0	50.4	52.7	55.6	60.4	64.0	65.0	65.1	62.7	59.9	55.4	51.6
		MCDB	52.1	54.9	60.1	66.3	73.2	82.0	84.5	84.3	76.9	67.5	58.2	54.0
Mean Daily Temperature Range	5% DB	MDBR	9.2	11.2	13.0	14.7	17.0	20.5	23.3	22.4	17.7	12.6	9.8	8.7
		MCDBR	11.4	13.5	15.4	16.9	19.8	21.7	22.7	22.0	19.7	16.2	11.6	10.2
		MCWBR	8.2	8.7	9.2	9.4	9.7	9.4	8.9	8.7	8.1	7.7	7.6	7.6
	5% WB	MCDBR	8.6	11.3	13.7	16.3	19.0	21.3	23.9	22.4	18.5	12.9	9.3	8.1
		MCWBR	7.4	8.1	8.9	9.5	10.0	9.5	9.5	9.2	8.1	7.1	7.2	7.3
Clear-Sky Solar Irradiance	taub	0.298	0.314	0.342	0.344	0.351	0.354	0.337	0.352	0.345	0.334	0.310	0.298	
	taud	2.523	2.481	2.409	2.416	2.422	2.429	2.447	2.409	2.454	2.512	2.532	2.530	
	Ebn at Noon	280	290	290	296	294	292	296	289	285	278	273	271	
	Edh at Noon	24	29	34	36	36	36	35	36	32	27	23	22	
All-Sky Solar Radiation	RadAvg	710	1014	1359	1713	2097	2397	2496	2193	1676	1123	777	627	
	RadStd	101	134	170	173	194	131	67	81	114	112	109	85	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218
Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page